Unconfounded but inflated causal estimates

Mathematical description

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I Setting

I. 1 Objectives

This document aims to provide a mathematical derivation to illustrate the trade-off underlined in this paper. Such a derivation may provide some intuition regarding the various mechanisms at play. It does not aim in any way to provide a general proof but merely a specific one, tied to the setting considered. While we keep this setting as general as possible, as in most cases, we have to make certain assumptions, in particular regarding the data generating process considered.

The objective here is to compare the exaggeration ratio for a naive regression with omitted variable bias (OVB) to that of a reduced form analysis of the impact of an exogenous shock on the explanatory variable of interest. We also compare it to the corresponding IV estimate. We expect the naive regression to be biased but more precise than the reduced form and the IV and therefore less subject to exaggeration problems. Roughly, we aim to derive the conditions under which the exaggeration due to the loss in precision is greater than the OVB. Thus, the objective is to compare the following quantities:

$$\mathbb{E}\left[\hat{\beta}_{\text{OVB}} |\ \beta_0, b_{\text{OVB}}, \sigma_{\text{OVB}}, \hat{\beta}_{\text{OVB}} > z_\alpha \sigma_{\text{OVB}}\right]$$
$$\mathbb{E}\left[\hat{\beta}_{\text{R}} |\ \beta_0, \sigma_{\text{R}}, \hat{\beta}_{\text{R}} > z_\alpha \sigma_{\text{R}}\right]$$
$$\mathbb{E}\left[\hat{\beta}_{\text{IV}} |\ \beta_0, \sigma_{\text{IV}}, \hat{\beta}_{\text{IV}} > z_\alpha \sigma_{\text{IV}}\right]$$

where OVB, R and IV denote respectively the naive regression, the reduced form and the IV. β represents the true effect of interest, $\hat{\beta}_{..}$ its estimate, $\sigma_{..}$ the standard error of the estimate, z_α the threshold value of a two-sided test of size α and $b_{..}$ the bias in the naive regression. Note that due to the exogeneity assumption $b_{\text{R}} = b_{\text{IV}} = 0$.

To compute these values, we need to know the distribution of the estimates. After describing the data generating process, we thus derive these asymptotic distribution of these estimates. We then compute and compare these quantities as a function of the parameters of interests. The main parameter of interest is linked the proportion of the variance in the explanatory variable of interest retained for identification.

I. 2 Data generating process

Consider the usual linear regression model to which we add omitted variables. We limit the number of covariates considered to a constant, an observed variable and an omitted variable. For all individual $i \in 1, ..., n$, we make the following assumptions:

$$y_i = \mathrm{x}_i' \boldsymbol{\beta} + \delta w_i + u_i \tag{1}$$

where $\mathrm{x}_i \in \mathbb{R}^2$ is a vector including a constant and the observed explanatory variable such that $\mathrm{x}_i = (1, x_i)'$, $y_i \in \mathbb{R}$ the observed outcome of interest, $w_i \in \mathbb{R}$ an unobserved omitted variable, $u_i \in \mathbb{R}$ an unobserved error term. $\boldsymbol{\beta} \in \mathbb{R}^2$ and $\delta \in \mathbb{R}$ are unknown parameters. $\boldsymbol{\beta} = (\beta_0, \beta_1)'$ and β_1 is the parameter of interest. For readability, we introduce the usual vector notation such that for instance $y = (y_1, ..., y_n)'$.

We further assume that the usual OLS assumptions hold: $(y_i, \mathrm{x}_i, w_i, u_i)$ are iid, finite second moments, *ie* $\mathbb{E}[y_i^2] < \infty$, $\mathbb{E}[w_i^2] < \infty$, $\mathbb{E}[x_i^2] < \infty$, positive-definiteness of $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$, u_i is conditional mean-zero, *ie* $\mathbb{E}[u_i|x_i] = 0$, and is uncorrelated with x_i, *ie* $\mathbb{E}[x_i u_i] = 0$. We also assume homoskedasticity such that $\mathbb{E}[u_i^2|x_i] = \sigma_u^2$ is constant.

We however assume that w_i is unobserved, correlated with x_i but not u_i, *ie* $\mathbb{E}[w_i u_i] = 0$, and that $\delta \neq 0$. To simplify the derivations, we further assume that the unobserved variable is centred, *ie* $\mathbb{E}[w_i] = 0$.

We further assume that there exists a valid instrumental variable z_i for x_i. Let $\mathrm{z} = (1, z_i)' \in \mathbb{R}^2$. Note that the model is just-identified. Since the instrument is valid, it satisfies exogeneity, *ie* $\mathbb{E}[\mathrm{z}_i u_i] = 0$ and $\mathbb{E}[z_i w_i] = 0$, relevance, *ie* $\mathrm{rank}(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']) = 2$, and positive-definiteness of $\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']$. To simplify interpretation, we further assume a data generating process for x_i:

$$x_i = \eta z_i + \gamma w_i + \epsilon_i \tag{2}$$

where $(\eta, \gamma) \in \mathbb{R}^2$ are unknown parameters. We assume that ϵ_i is uncorrelated with z_i and w_i, *ie* $\mathbb{E}[z_i \epsilon_i] = 0$ and $\mathbb{E}[w_i \epsilon_i] = 0$. Note that, since x_i and w_i are correlated, $\gamma \neq 0$.

The following Directed Acyclic Graph (DAG) sums up the relationships between the variables. It also helps underline the independence, or not, of some variables.

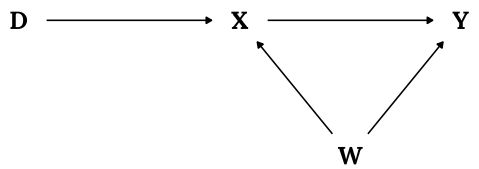

Figure 1 – DAG of the data generating process

II Asymptotic distribution of the estimates

As discussed above, the goal is to compare the exaggeration ration in the case of the naive regression to that of the corresponding reduced form and IV. In this section, we **derive the asymptotic distributions of these estimates**.

II. 1 Naive regression (OVB)

The true DGP is given by equation 1. Since, we do not observe W, we consider the projection of y on X only:

$$y = X\boldsymbol{\beta}_{\text{OVB}} + v \tag{3}$$

where $v = (v_1, ..., v_n)' \in \mathbb{R}^n$ and by definition of the projection, $\mathbb{E}[X'v] = 0$.

II. 1.1 Bias of the estimator

From equation 3 we get:

$$\begin{aligned}
& X'y = X'X\boldsymbol{\beta}_{\text{OVB}} + X'v \\
\Rightarrow\quad & \mathbb{E}[X'y] = \underbrace{\mathbb{E}[X'X]}_{\text{pos. def.}}\boldsymbol{\beta}_{\text{OVB}} + \underbrace{\mathbb{E}[X'v]}_{0} \\
\Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \mathbb{E}[X'X]^{-1}\mathbb{E}[X'y] \\
\Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \mathbb{E}[X'X]^{-1}\mathbb{E}[X'(X\boldsymbol{\beta} + \delta w + u)] && \text{cf eq. 1} \\
\Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \boldsymbol{\beta} + \mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]\delta + \underbrace{\mathbb{E}[X'u]}_{0} \\
\Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \boldsymbol{\beta} + \mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]\delta
\end{aligned}$$

Since X and w are correlated and $\delta \neq 0$, $\boxed{\boldsymbol{\beta}_{\text{OVB}} \neq \boldsymbol{\beta}}$.

Let $\hat{\boldsymbol{\beta}}_{\text{OVB}}$ denote the least square estimator of $\boldsymbol{\beta}_{\text{OVB}}$. We thus have:

$$\hat{\boldsymbol{\beta}}_{\text{OVB}} = (X'X)^{-1}X'y \tag{4}$$

$$\begin{aligned}
\Rightarrow\quad & \hat{\boldsymbol{\beta}}_{\text{OVB}} = (X'X)^{-1}X'(X\boldsymbol{\beta}_{\text{OVB}} + v) && \text{cf eq. 3} \\
\Rightarrow\quad & \mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}} \mid X] = \boldsymbol{\beta}_{\text{OVB}} + (X'X)^{-1}\underbrace{\mathbb{E}[X'v]}_{0} \\
\Rightarrow\quad & \mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}}] = \mathbb{E}[\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}} \mid X]] = \boldsymbol{\beta}_{\text{OVB}} && \text{by the Law of Iterated Expect. (LIE)}
\end{aligned}$$

Thus, $\hat{\boldsymbol{\beta}}_{\text{OVB}}$ is an unbiased estimator of $\boldsymbol{\beta}_{\text{OVB}}$ but a biased estimator of $\boldsymbol{\beta}$. We then compute its asymptotic distribution.

II. 1.2 Asymptotic distribution

We can write:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) = \left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i\mathrm{x}_i'\right)^{-1} \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i v_i\right)$$

By the Weak Law of Large Numbers (WLLN), $\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i\mathrm{x}_i' \xrightarrow{p} \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$.
Since $\mathbb{E}[\mathrm{x}_i v_i] = 0$, by the Central Limit Theorem (CLT), $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i v_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2])$.
Applying Slutsky's theorem then yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\right)$$

And thus,

$$\hat{\boldsymbol{\beta}}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta} + \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i]\delta, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\right)$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}$, $\mathbb{E}[\mathrm{x}_i w_i]$ and $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]$.

$$\mathbb{E}[\mathrm{x}_i\mathrm{x}_i'] = \mathbb{E}\begin{bmatrix} 1 & x \\ x & x^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_x \\ \mu_x & \sigma_x^2 + \mu_x^2 \end{bmatrix} \quad \Rightarrow \quad \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1} = \frac{1}{\sigma_x^2}\mathbb{E}\begin{bmatrix} \sigma_x^2 + \mu_x^2 & -\mu_x \\ -\mu_x & 1 \end{bmatrix}$$

$$\mathbb{E}[\mathrm{x}_i w_i] = \mathbb{E}\begin{bmatrix} w_i \\ x_i w_i \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbb{E}[x_i]\underbrace{\mathbb{E}[w_i]}_{0} + \mathrm{cov}(x_i, w_i) \end{bmatrix} = \begin{bmatrix} 0 \\ \underbrace{\mathrm{cov}(z_i, w_i)}_{0} + \gamma\underbrace{\mathrm{cov}(w_i, w_i)}_{\sigma_w^2} + \underbrace{\mathrm{cov}(\epsilon_i, w_i)}_{0} \end{bmatrix} = \begin{bmatrix} 0 \\ \gamma\sigma_w^2 \end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i] = \frac{\gamma\sigma_w^2}{\sigma_x^2}\begin{bmatrix} -\mu_x \\ 1 \end{bmatrix}$$

Note that $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2] \overset{\text{LIE}}{=} \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\mathbb{E}[v_i^2|\mathrm{x}_i]]$. We thus first compute $\mathbb{E}[v_i^2|\mathrm{x}_i]$, noting that:

$$\begin{aligned}
v_i &= y_i - \mathrm{x}_i'\boldsymbol{\beta}_{\text{OVB}} \\
&= \mathrm{x}_i'(\boldsymbol{\beta} - \boldsymbol{\beta}_{\text{OVB}}) + \delta w_i + u_i \\
&= -\mathrm{x}_i(\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i]\delta) + \delta w_i + u_i \\
&= -\frac{\delta\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x) + \delta w_i + u_i \\
\Rightarrow \quad v_i^2 &= \left(\delta w_i + u_i - \frac{\delta\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\right)^2 \\
&= \delta^2 w_i^2 + u_i^2 + \left(\frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)^2 (x_i - \mu_x)^2 + 2\delta w_i u_i + 2\left(\frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)(x_i - \mu_x)(\delta w_i + u_i) \\
\Rightarrow \quad \mathbb{E}[v_i^2|\mathrm{x}_i] &= \delta^2\mathbb{E}[w_i^2|\mathrm{x}_i] + \underbrace{\mathbb{E}[u_i^2|\mathrm{x}_i]}_{\sigma_u^2} + \left(\frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)^2 \underbrace{\mathbb{E}[(x_i - \mu_x)^2|\mathrm{x}_i]} \\
&\quad + 2\delta\underbrace{\mathbb{E}[w_i u_i|\mathrm{x}_i]} + 2\left(\frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)(x_i - \mu_x)(\delta\underbrace{\mathbb{E}[w_i|\mathrm{x}_i]} + \underbrace{\mathbb{E}[u_i|\mathrm{x}_i]}_{0})
\end{aligned}$$

Thus, for $\hat{\beta}_{\text{OVB}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$, we have:

$$\boxed{\hat{\beta}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2}, \ \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n\ \sigma_x^2}\right)} \tag{5}$$

Using the notations introduced in section I. 1, we have:

$$\begin{aligned}
b_{\text{OVB}} &= \frac{\delta\gamma\sigma_w^2}{\sigma_x^2} \\
\sigma_{\text{OVB}} &= \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n\ \sigma_x^2}
\end{aligned}$$

Now, note that we have :

$$\begin{aligned}\sigma_x^2 &= \mathrm{Var}(x) = \mathrm{Var}(z_i + \gamma w_i + \epsilon_i)\\ &= \mathrm{Var}(z_i) + \gamma^2 \mathrm{Var}(w_i) + \mathrm{Var}(\epsilon_i) + 2\gamma \underbrace{\mathrm{cov}(z_i, w_i)}_{0} + 2\gamma \underbrace{\mathrm{cov}(\epsilon_i, w_i)}_{0} + 2 \underbrace{\mathrm{cov}(\epsilon_i, z_i)}_{0}\\ &= \sigma_z^2 + \gamma^2 \sigma_w^2 + \sigma_\epsilon^2\end{aligned}$$

II. 2 Instrumental Variable

The true DGP for y is given by equation 1. Here, we instrument x_i with z_i, or more precisely $\mathrm{x}_i = (1, x_i)'$ with $\mathrm{z}_i = (1, z_i)'$. We are therefore in a just-identified case and $\hat{\boldsymbol{\beta}}_{\text{2SLS}} = \hat{\boldsymbol{\beta}}_{\text{IV}}$. And:

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{\text{IV}} &= (Z'X)^{-1} Z' y\\ &= (Z'X)^{-1} Z'(X\boldsymbol{\beta} + \delta w + u)\\ &= \boldsymbol{\beta} + (Z'X)^{-1} Z' w \delta + (Z'X)^{-1} Z' u\end{aligned}$$

Thus,

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{IV}} | X, Z] = \boldsymbol{\beta} + (Z'X)^{-1} \underbrace{\mathbb{E}[Z'w]}_{0} \delta + (Z'X)^{-1} \underbrace{\mathbb{E}[Z'u]}_{0} = \boldsymbol{\beta}$$

$\hat{\boldsymbol{\beta}}_{\text{IV}}$ is therefore an unbiased estimator of $\boldsymbol{\beta}$. We then derive its asymptotic distribution, noting that:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{IV}} - \boldsymbol{\beta}) = \left(\frac{1}{n}\sum_{i=1}^n \mathrm{z}_i \mathrm{x}_i'\right)^{-1} \delta \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \mathrm{z}_i w_i\right) + \left(\frac{1}{n}\sum_{i=1}^n \mathrm{z}_i \mathrm{x}_i'\right)^{-1} \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \mathrm{z}_i u_i\right)$$

The WLLN gives $\frac{1}{n}\sum_{i=1}^n \mathrm{z}_i \mathrm{x}_i' \xrightarrow{p} \mathbb{E}[\mathrm{z}_i \mathrm{x}_i']$ and since $\mathbb{E}[\mathrm{z}_i w_i] = 0$ and $\mathbb{E}[\mathrm{z}_i u_i] = 0$, the Central Limit Theorem (CLT) gives $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \mathrm{z}_i w_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' w_i^2])$ and $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \mathrm{z}_i u_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' u_i^2])$.

Now, $\mathbb{E}[\mathrm{z}_i \mathrm{z}_i' w_i^2] = \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' \mathbb{E}[w_i^2 | \mathrm{z}_i]] = \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' \mathbb{E}[w_i^2]] = \sigma_w^2 \mathbb{E}[\mathrm{z}_i \mathrm{z}_i']$ and similarly, $\mathbb{E}[\mathrm{z}_i \mathrm{z}_i' u_i^2] = \sigma_u^2 \mathbb{E}[\mathrm{z}_i \mathrm{z}_i']$ since

III Exaggeration ratio and its properties